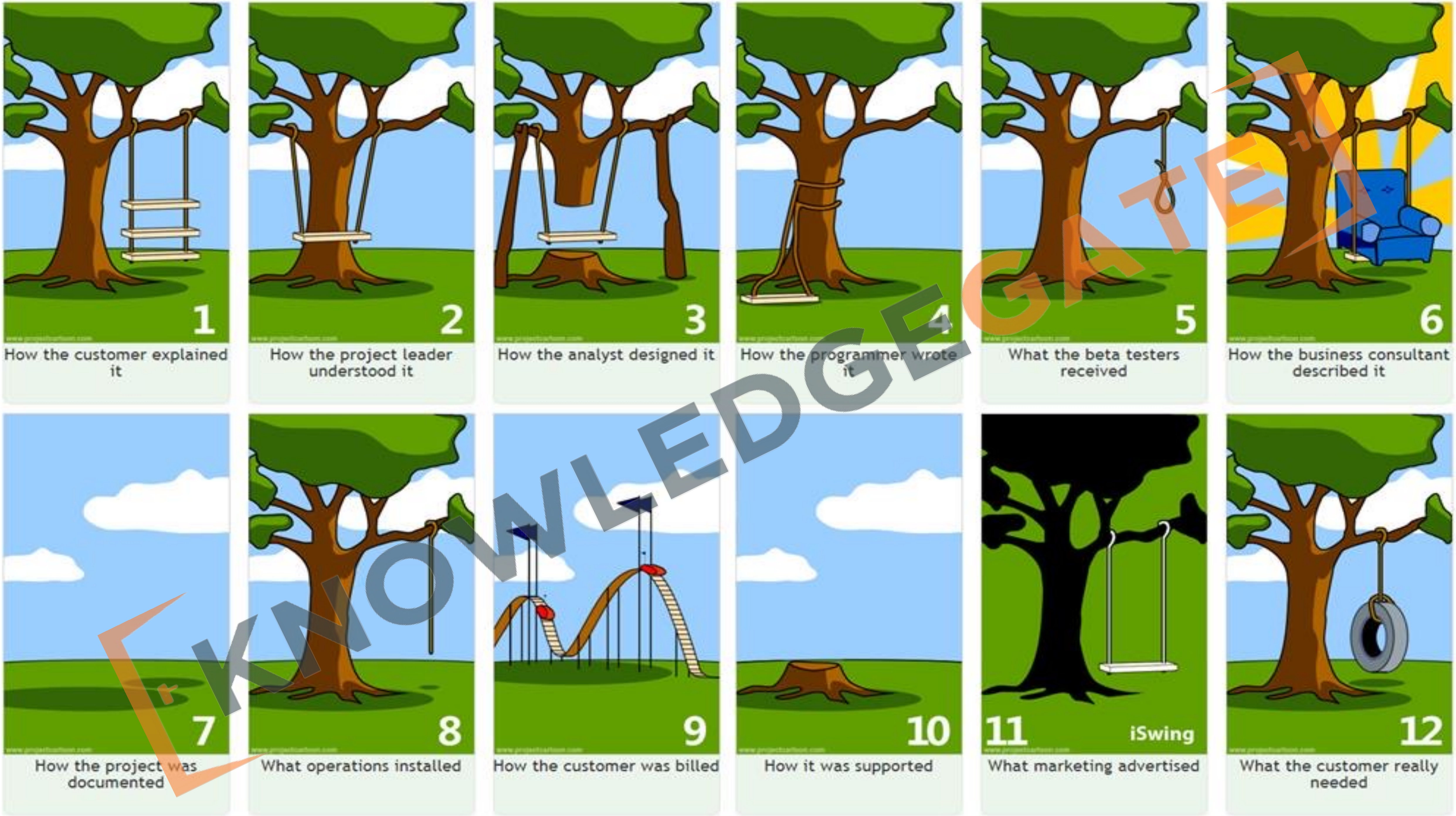
Requirement Analysis

- The hardest part of building a software is deciding precisely what is to be built. Requirement engineering is the disciplined application of proven principle, methods, tools and notations to describe a proposed system's intended behaviour and its associated constraints.


1
How the customer explained it
2
How the project leader understood it
3
How the analyst designed it
4
How the programmer wrote it
5
What the beta testers received
6
How the business consultant described it
7
How the project was documented
8
What operations installed
9
How the customer was billed
10
How it was supported
11
iSwing
What marketing advertised
12
What the customer really needed

Difficulties of Requirement Engineering

- **Requirement are difficult to uncover**: no one give the complete requirement in the first time, if someone does then also requirement are incomplete

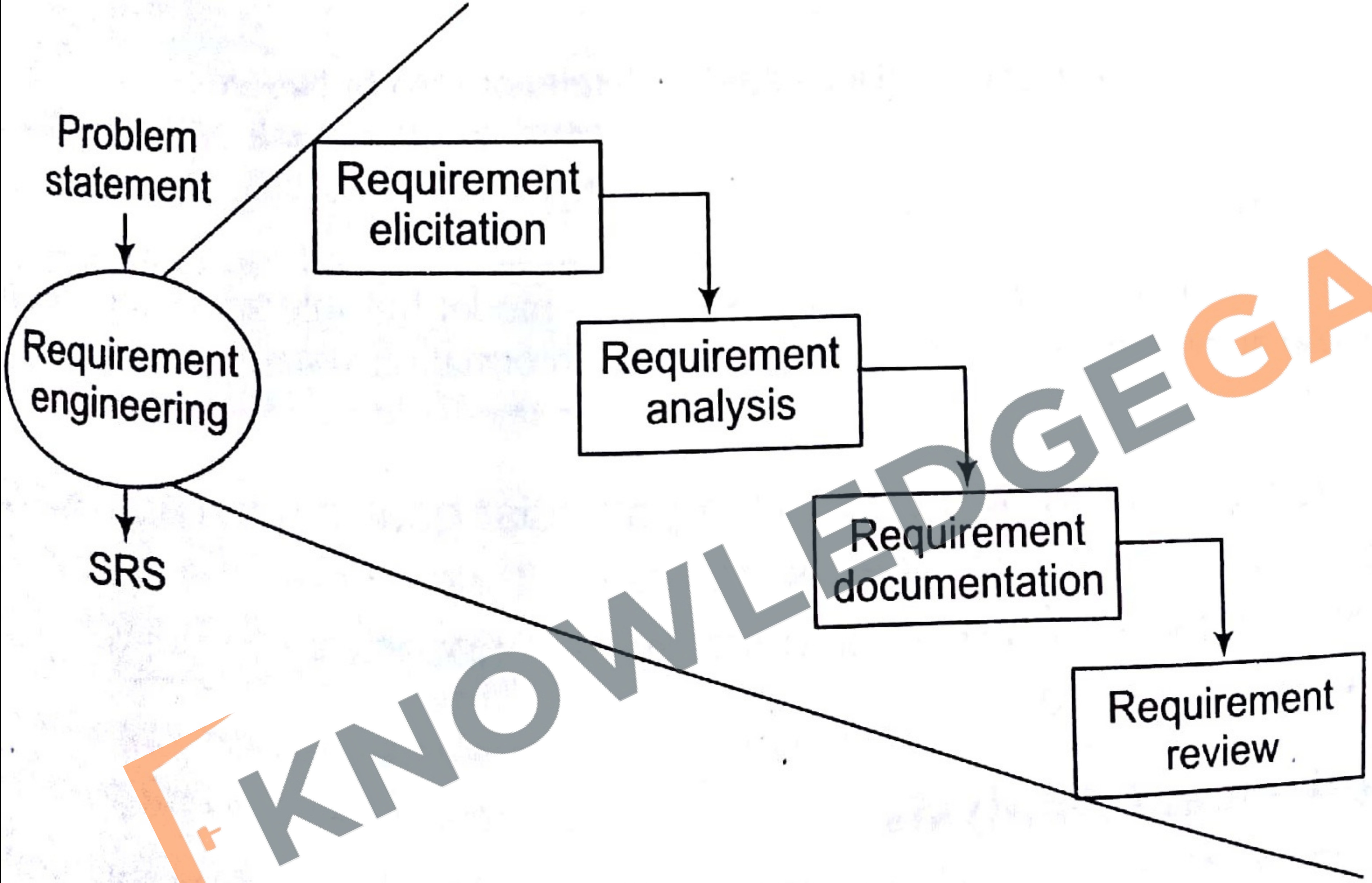
- **Requirement Changes**: with the development process requirements get added and changed as the user begins to understand the system and his or her real need

- **<u>Tight project schedule</u>**: have insufficient time to do a decent job

- **<u>Communication Barrier</u>**: user and developer have different technical background and tastes

- **Lack of resources**: there may not be enough resources to build software that can do everything the customer wants

- Types of requirement
  - **Known requirement**: Which is already known to the stake holder
  - **Unknown Requirement**: Forgotten by stake holder because that are not needed right now
  - **Undreamt Requirement**: Stakeholder is unable to think of new requirement due to limited domain knowledge



Various steps of analysis and specifications

- There are majorly four steps in requirement analysis
    - **Requirement Elicitation**: Gathering of Requirement
    - **Requirement Analysis**: Requirements are analysed to identify inconsistences, defects etc and to resolve conflicts
    - **Requirement documentation**: Here we document all the refined requirement properly
    - **Requirement Review**: Review is carried out to improve the quality of the SRS

Methods of requirement elicitation

- Is the most difficult, most critical, most error-prone and communication intensive aspect of the s/w development. It can only succeed only through an effective customer developer partnership.

- **Interview**
  - Both parties would like to understand each other
  - Interview can be of two types open-ended or structured
  - In open ended, there is no present agenda, context free questions can be asked to understand the problem, to have an over view over the situation
  - In structured interview, agenda is pre-set.

- **<u>Brain Storming</u>**
  - A kind of group discussion, which lead to ideas very quickly and help to promote creative thinking
  - Very popular now a days and is being used in most of the organizations
  - All participants are encouraged to say whatever idea come to their mind and no one will be criticized for any idea no matter how goofy it seems

- **Delphi technique**
  - Here participants are made to write the requirement on a piece of paper, then these requirements are exchanged among participants who gave their comments to get a revised set of requirements. This process is repeated till the final consensus is reached

- **<u>FAST (facilitated application specification technique)</u>**
  - This approach encourages the creation of joint team of customer and developer who works together to understand correct set of requirements
  - Everyone is asked to prepare a list of
    - What surrounds the system
    - Produced by the system
    - Used by the system
    - List of service, constraints, and performance criterion
    - Then we divide these lists into smaller list to work in smaller teams

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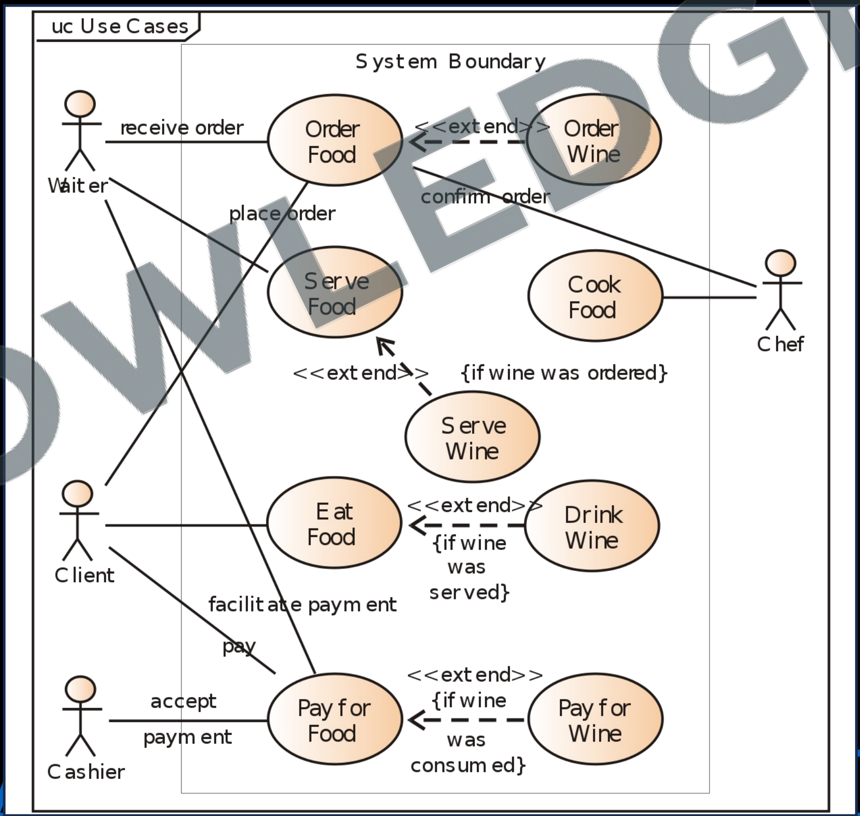
- <u>QFD (quality functional deployment)</u>
  - Its emphases to incorporate the voice of the customer with importance
  - Then according to customer, a value indicating a degree of importance, is assigned to each requirement. Thus, the customer, determine the importance of each requirement on a scale of 1 to 5 as:
    - **5 points**: very important
    - **4 points**: important
    - **3 points**: not important but nice to have
    - **2 point**: not important
    - **1 point**: unrealistic, requires further explanation

- **<u>Use case approach</u>**
  - These are structured description of the user requirement. It is a narrative which describe the sequence of events from user's perspective
  - Use case diagrams are graphical representation to show the system at different levels
  - They are some times supportive my the activity diagrams, to understand the work flow



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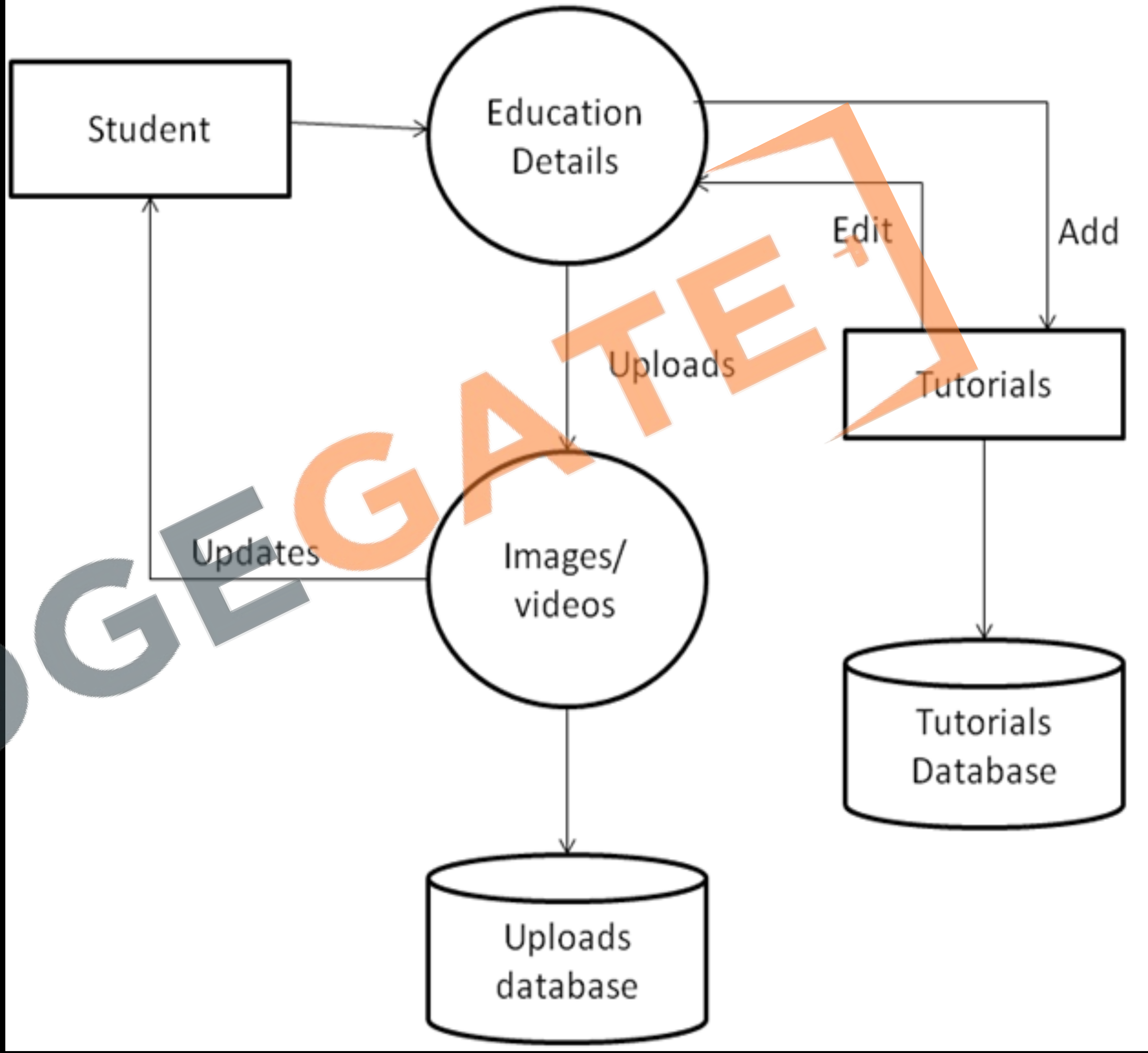
Requirement analysis

- In this phase we analysis all the set of requirements to find any inconsistency or conflicts.
- In requirement gathering phase our all concentration was on getting all the set of requirements but now, we see how many requirements are contradictory to each other or requires further exploration to be considered further.
- Different tools can be used Data flow diagram, Control flow diagram, ER diagram etc

Data Flow Diagram

- A data flow diagram or bubble chart is a graphical representation of the flow of data through a system. It clarifies systems requirements and identify major transformations.

- DFD represent a system at different level of abstraction. DFD may be Partitioned into levels that represent increasing information flow and functional details



- Components of DFD
  - Function/Process
  - Data Store
  - External Entity
  - Data Flow

DFD uses hierarchy to maintain transparency thus multilevel DFD's can be created. Levels of DFD are as follows:

- **<u>0-level DFD</u>**: It represents the entire system as a single bubble and provides an overall picture of the system.

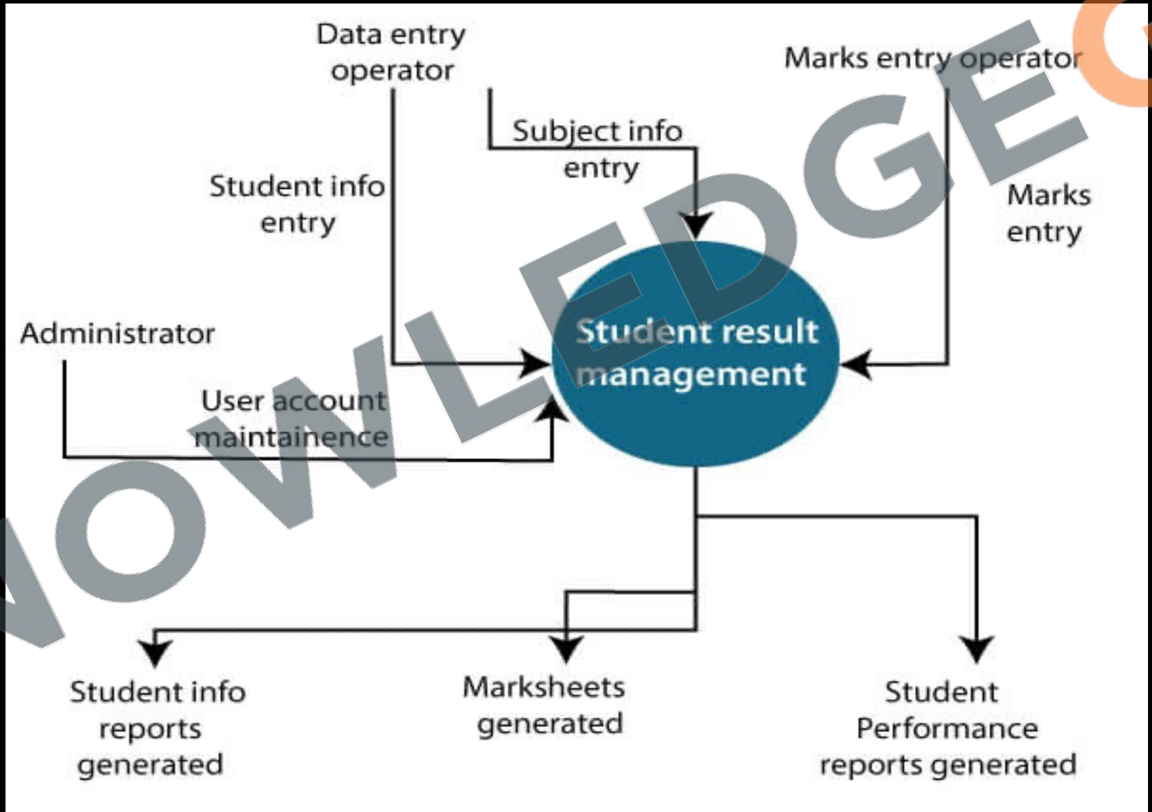


Fig: Level-0 DFD of result management system

- **1-level DFD**: It represents the main functions of the system and how they interact with each other.

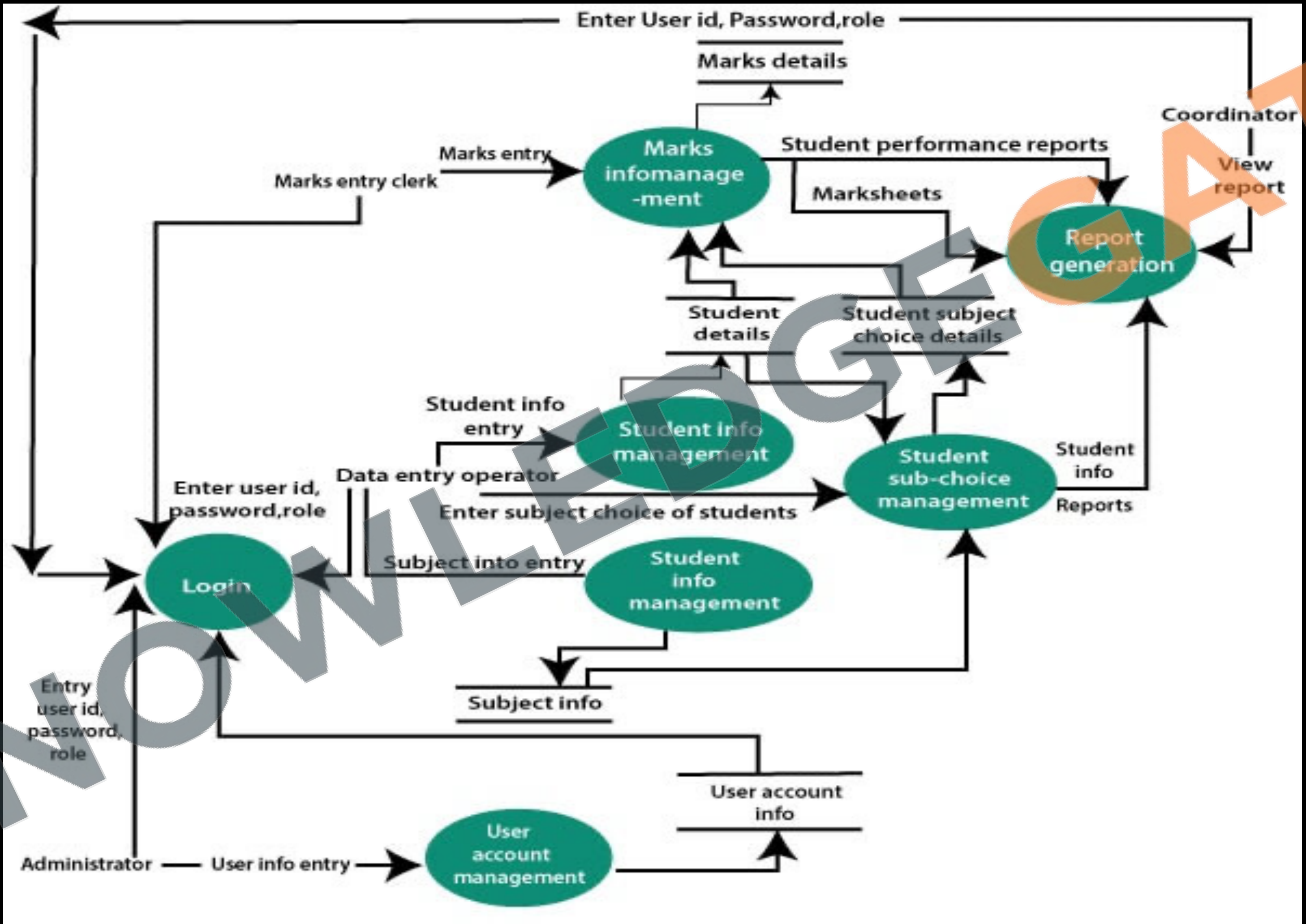


Fig: Level-1 DFD of result management system

- **2-level DFD**: It represents the processes within each function of the system and how they interact with each other.

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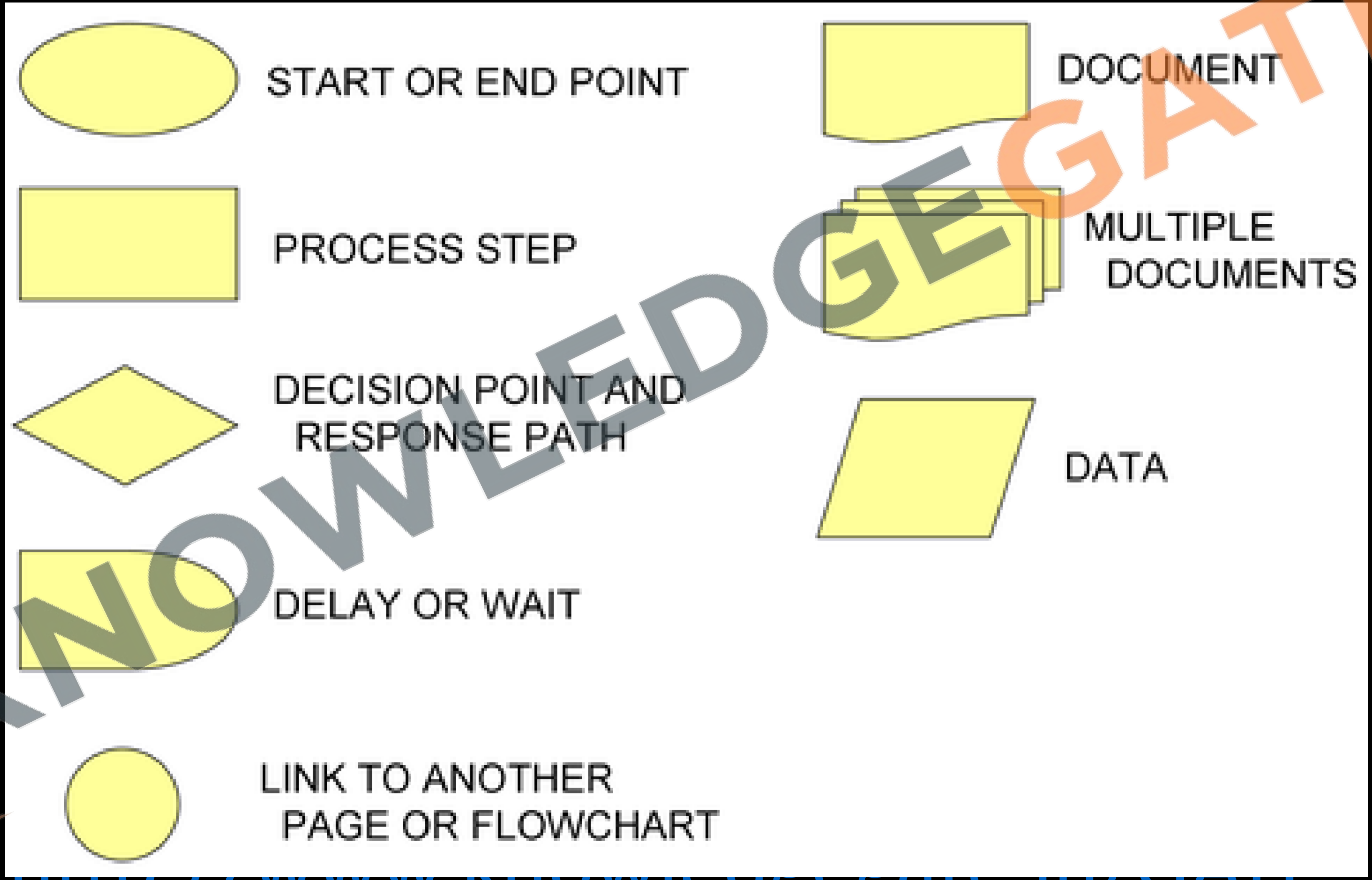
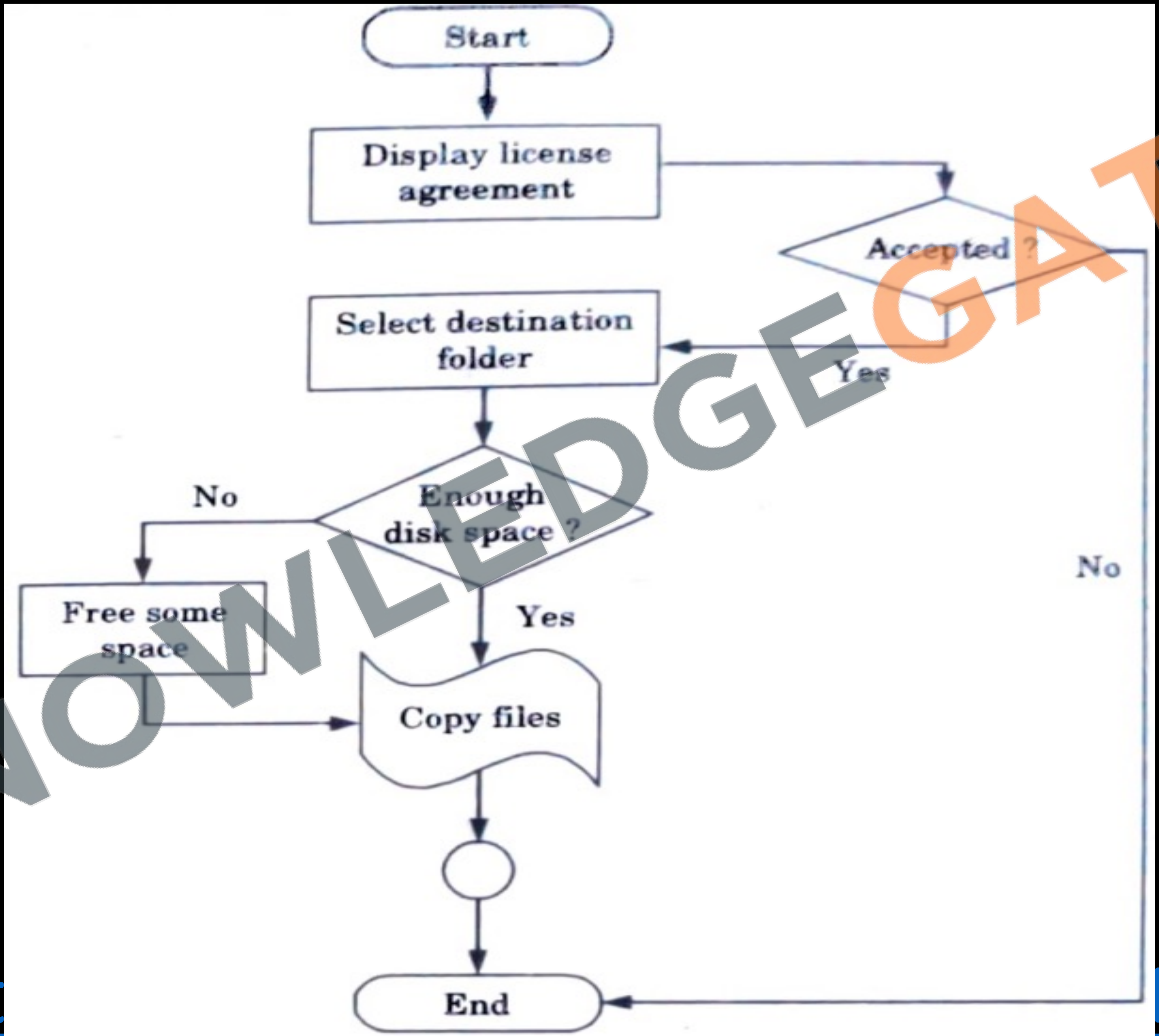
Control Flow Diagram/Control flow chart/Flow Chart

- A flow chart is a graphical representation of how control flow during the execution of a program. It use the following symbols to represent a system’s control flow.



- Flow Chart of software installation



Data Dictionary

- **Purpose**: Serves as a repository for data item details in Data Flow Diagrams, ensuring consistent definitions between customers and developers.
- **Content**: Includes data item name, aliases, and purpose, promoting clear understanding.
- **Relationships & Range**: Records data item relationships and value ranges (e.g., marks between 0-100).
- **Data Flows & Structures**: Tracks process interactions with data items and documents their structure or composition.
- **Role in Requirements**: Essential in initial stages for defining customer data items, aligning developer and customer understanding.

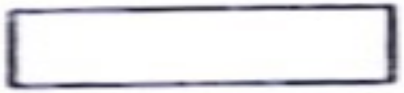
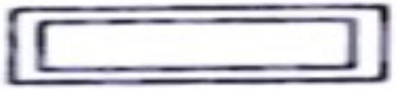
Data Dictionary

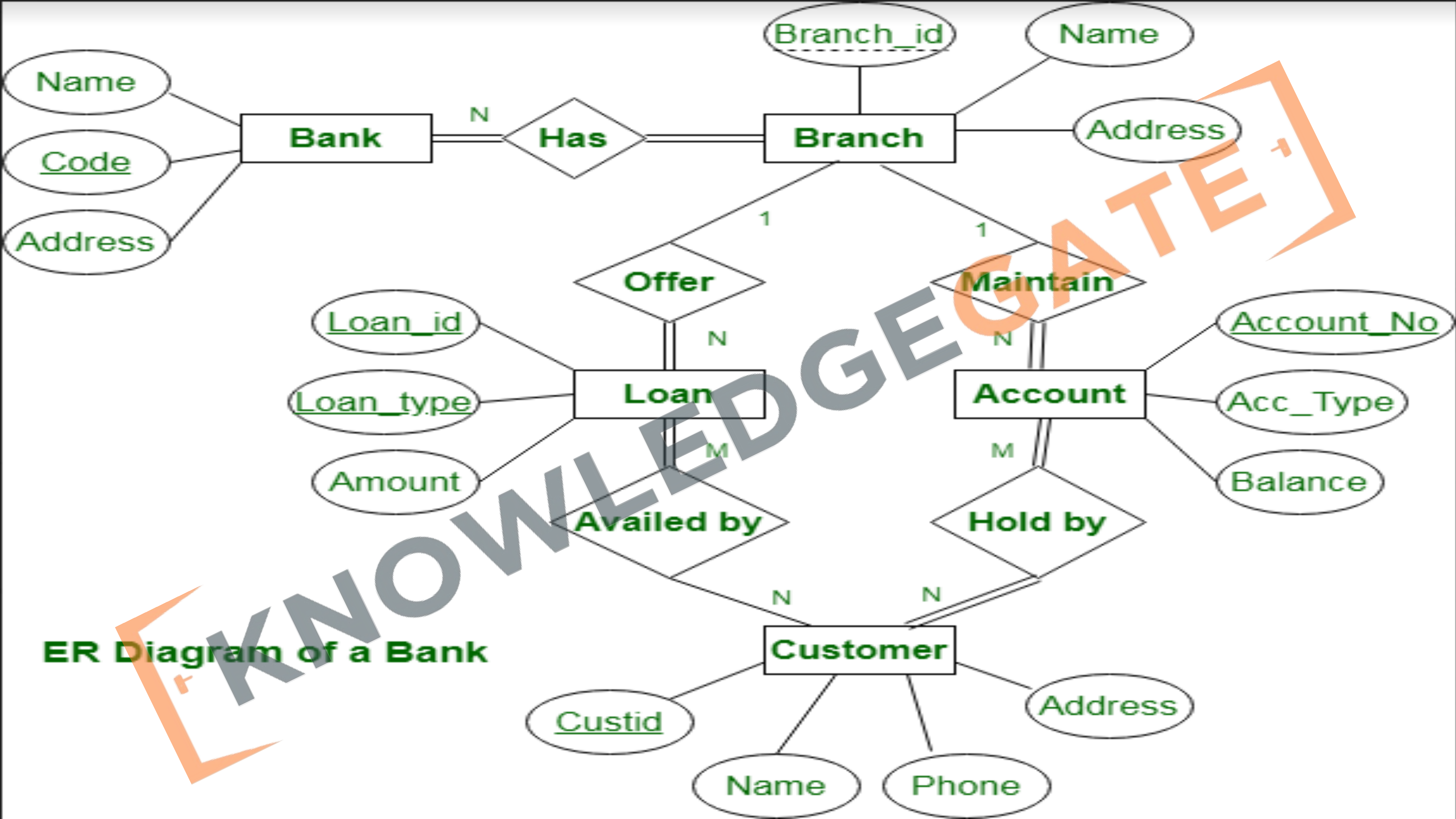
Data Dictionary outlining a Online Gaming Service

Data Item	Data Type	Data Format	Number of Bytes for Storage	Size for Display	Description	Example	Validation
MemberID	String	XNNNNNN	7	7	Unique Identifier For Member	M123456	
First Name	String		25	25	First Name of Member	Scott	
Surname	String		25	25	Last Name of Member	Daniels	
D.O.B	Floating Point (Date Format)	DD/MM/YYYY	4	10	Birth Date of Member	02/04/1990	Date < Today - 15 years
Platinum Membership?	Boolean	X	1	1	True (T) or False (F)	T	
Subscription Cost	Floating Point (Currency Format)	$NN.NN	4	6	Cost of Members Subscription	$27.50	Cost > 0 Cost < $50.00
Available Game Packs	Array (String)		25 * Number of Games	25 * Number of Games	Names of Game Packs	Open World Game Pack	

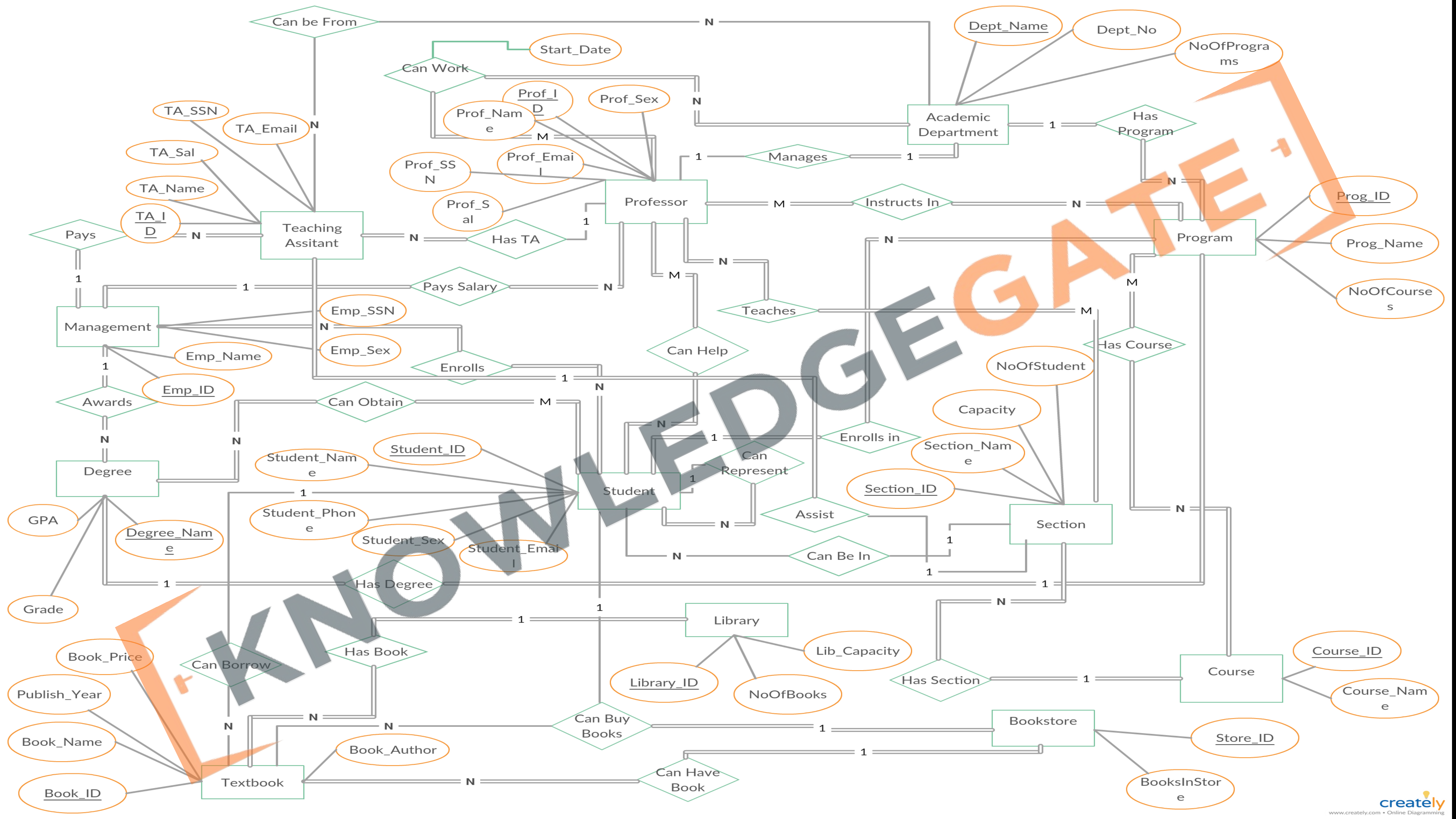
Entity Relationship Diagram

- ER Diagram a non-technical design method works on conceptual level based on the perception of the real world.
- E-R data model was developed to facilitate software designers by allowing specification of an enterprise schema that represents *the overall logical structure of a database*.
- Three main constructs are data entities, their relationships and their associated attributes

S. No.	Component name	Symbol	Description
1.	Rectangles		Represents entity sets
2.	Ellipses		Represents attributes
3.	Diamonds		Represents relationship sets
4.	Lines		Links attributes to entity sets and entity sets to relationship sets
5.	Double ellipses		Represents multivalued attributes
6.	Dashed ellipses		Represents derived attributes
7.	Double rectangles		Represents weak entity sets
8.	Double lines		Represents total participation of an entity in a relationship set



ER Diagram of a Bank



Can be From
Dept_Name
Dept_No
NoOfPrograms
Can Work
Start_Date
Prof_ID
Prof_Sex
Prof_Name
Academic Department
Has Program
TA_SSN
TA_Email
TA_Sal
Manages
Prof_SSN
Prof_Email
TA_Name
Prof_Sal
Professor
Instructs In
Prog_ID
TA_ID
Pays
Teaching Assitant
Has TA
Program
Prog_Name
Pays Salary
NoOfCourses
Management
Emp_SSN
Teaches
Emp_Sex
Emp_Name
Can Help
Has Course
Enrolls
NoOfStudent
Emp_ID
Awards
Can Obtain
Capacity
Enrolls in
Student_ID
Section_Name
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Student
Student_Phone
Assist
Section
GPA
Degree_Name
Student_Sex
Student_Email
Can Be In
Has Degree
Grade
Library
Can Borrow
Has Book
Lib_Capacity
Book_Price
Has Section
Course
Course_ID
Publish_Year
Library_ID
NoOfBooks
Course_Name
Can Buy Books
Bookstore
Store_ID
Book_Name
Book_Author
Can Have Book
BooksInStore
Book_ID
Textbook

Decision Tables

- A decision table is a brief visual representation for specifying which actions to perform depending on given conditions.
- A decision table is a good way to settle with different combination inputs with their corresponding outputs.
- Decision tables are very much helpful in requirements management and test design techniques. It provides a regular way of starting complex business rules, that is helpful for developers as well as for testers.

	Conditions/ Courses of Action	Rules					
		1	2	3	4	5	6
Condition Stubs	Employee type	S	H	S	H	S	H
	Hours worked	<40	<40	40	40	>40	>40
Action Stubs	Pay base salary	X		X		X	
	Calculate hourly wage		X		X		X
	Calculate overtime						X
	Produce Absence Report		X				

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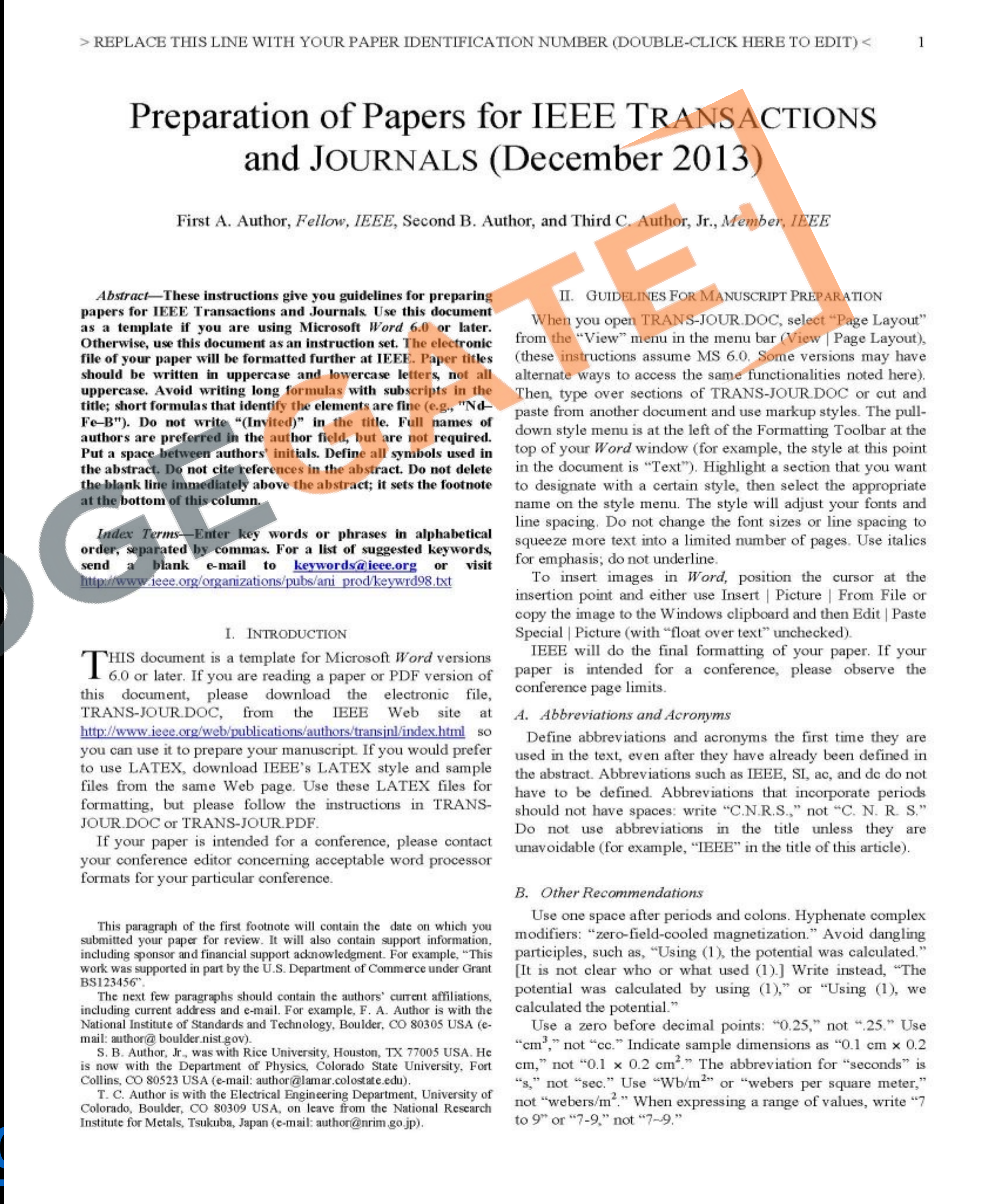
Aspect	Decision Table	Decision Tree
Structure	Tabular format, lists conditions, actions, and rules.	Tree-like diagram, shows decision paths with branches.
Decision Making	Based on various combinations of conditions leading to specific actions.	Based on sequential decisions leading to outcomes.
Ease of Use	Simple for representing complex logical relationships, can become cumbersome with many variables.	More intuitive for visualizing decision paths, can become complex with many branches.
Application	Preferred in scenarios where all possible combinations of conditions and outcomes need clear representation.	Ideal for situations where decision-making process is more sequential and hierarchical.

Requirement Documentation

- After have final set of requirements it is also necessary to document them properly in a standard format, so that can be understood easily even by non-technical person.
- Here IEEE provides standard for SRS documents in IEEE830, using which we can make SRS document more readable, modifiable, and a format which can be followed through the world.



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Preparation of Papers for IEEE TRANSACTIONS and JOURNALS (December 2013)

First A. Author, *Fellow, IEEE*, Second B. Author, and Third C. Author, Jr., *Member, IEEE*

***Abstract*—These instructions give you guidelines for preparing papers for IEEE Transactions and Journals. Use this document as a template if you are using Microsoft *Word* 6.0 or later. Otherwise, use this document as an instruction set. The electronic file of your paper will be formatted further at IEEE. Paper titles should be written in uppercase and lowercase letters, not all uppercase. Avoid writing long formulas with subscripts in the title; short formulas that identify the elements are fine (e.g., "Nd–Fe–B"). Do not write "(Invited)" in the title. Full names of authors are preferred in the author field, but are not required. Put a space between authors' initials. Define all symbols used in the abstract. Do not cite references in the abstract. Do not delete the blank line immediately above the abstract; it sets the footnote at the bottom of this column.**

***Index Terms*—Enter key words or phrases in alphabetical order, separated by commas. For a list of suggested keywords, send a blank e-mail to keywords@ieee.org or visit** http://www.ieee.org/organizations/pubs/ani_prod/keywrd98.txt

I. INTRODUCTION

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If your paper is intended for a conference, please contact your conference editor concerning acceptable word processor formats for your particular conference.

This paragraph of the first footnote will contain the date on which you submitted your paper for review. It will also contain support information, including sponsor and financial support acknowledgment. For example, "This work was supported in part by the U.S. Department of Commerce under Grant BS123456".

The next few paragraphs should contain the authors' current affiliations, including current address and e-mail. For example, F. A. Author is with the National Institute of Standards and Technology, Boulder, CO 80305 USA (e-mail: author@ boulder.nist.gov).

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II. GUIDELINES FOR MANUSCRIPT PREPARATION

When you open TRANS-JOUR.DOC, select "Page Layout" from the "View" menu in the menu bar (View | Page Layout), (these instructions assume MS 6.0. Some versions may have alternate ways to access the same functionalities noted here). Then, type over sections of TRANS-JOUR.DOC or cut and paste from another document and use markup styles. The pull-down style menu is at the left of the Formatting Toolbar at the top of your *Word* window (for example, the style at this point in the document is "Text"). Highlight a section that you want to designate with a certain style, then select the appropriate name on the style menu. The style will adjust your fonts and line spacing. Do not change the font sizes or line spacing to squeeze more text into a limited number of pages. Use italics for emphasis; do not underline.

To insert images in *Word*, position the cursor at the insertion point and either use Insert | Picture | From File or copy the image to the Windows clipboard and then Edit | Paste Special | Picture (with "float over text" unchecked).

IEEE will do the final formatting of your paper. If your paper is intended for a conference, please observe the conference page limits.

A. Abbreviations and Acronyms

Define abbreviations and acronyms the first time they are used in the text, even after they have already been defined in the abstract. Abbreviations such as IEEE, SI, ac, and dc do not have to be defined. Abbreviations that incorporate periods should not have spaces: write "C.N.R.S.," not "C. N. R. S." Do not use abbreviations in the title unless they are unavoidable (for example, "IEEE" in the title of this article).

B. Other Recommendations

Use one space after periods and colons. Hyphenate complex modifiers: "zero-field-cooled magnetization." Avoid dangling participles, such as, "Using (1), the potential was calculated." [It is not clear who or what used (1).] Write instead, "The potential was calculated by using (1)," or "Using (1), we calculated the potential."

Use a zero before decimal points: "0.25," not ".25." Use "cm^3," not "cc." Indicate sample dimensions as "0.1 cm × 0.2 cm," not "0.1 × 0.2 cm^2." The abbreviation for "seconds" is "s," not "sec." Use "Wb/m^2" or "webers per square meter," not "webers/m^2." When expressing a range of values, write "7 to 9" or "7-9," not "7~9."

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IEEE Standard for SRS

1-**Introduction**

1.1-Purpose

1.2-Scope/Intended Audience

1.3-Definition, Acronyms and Abbreviation

1.4-References / Contact Information / SRS team member

1.5-Overview

2-**Overall Description**

- 2.1-Product Perspective
- 2.2-Product Functions
- 2.3-User Characteristics
- 2.4-General Constraints
- 2.5-Assumptions and dependencies

3-**Specific Requirement**

- 3.1-External interface requirements(user/hardware/software interface)
- 3.2-Functional requirements
- 3.3-Performance requirements
- 3.4-Design constraints(Standards compliance/hardware limitations)
- 3.5-Logical database requirements
- 3.6-Software system attributes(Reliability, availability, Security, Maintainability)

4-**Change Management process**

5-**Document approval**

5.1-Tables, diagrams, and flowcharts

5.2-Appendices

5.3-Index

Requirement Review

- Before finalising the requirement one more review specially by a third party, who a master in the industry is advisable, to have a fresh look over the system, and can mentions any points if missed by team.